

M(1st Sm.)-Computer Sc.-H/CC-1/(CBCS)

(2)

4. (a) Design a BCD-to-Excess 3 code converter using minimum number of NAND gates.
(b) Write down the procedure to convert a given AND-OR gate network to all NAND gates network. 8+2
5. (a) Implement the logic expression $f = \sum m(0, 2, 5, 7)$ by 1:8 demultiplexer.
(b) Design a two-bit comparator by discrete logic gates. 4+6
6. (a) Design S-R flip-flop using NAND gates only.
(b) Convert it to J-K flip-flop.
(c) Can a $\bar{D}$ flip-flop be made to operate in the toggle mode? 4+4+2
7. (a) Design a counter with counting sequence 0-3-5-1-7-6-0... using J-K flip-flops and other logic gates.
(b) Can a counter be made to start count from any pre-defined value?— Explain. 8+2
8. (a) Design an asynchronous MOD-4 DOWN counter.
(b) Find the outputs of the key diode matrix circuit in Figure (1), where the input keys are pressed one at a time, K_0, K_1, K_2 are press to ON switch, D_0 to D_6 are PN Junction diode, R_0 to R_3 are resistors and Y_0 to Y_3 are outputs. Justify your answer.

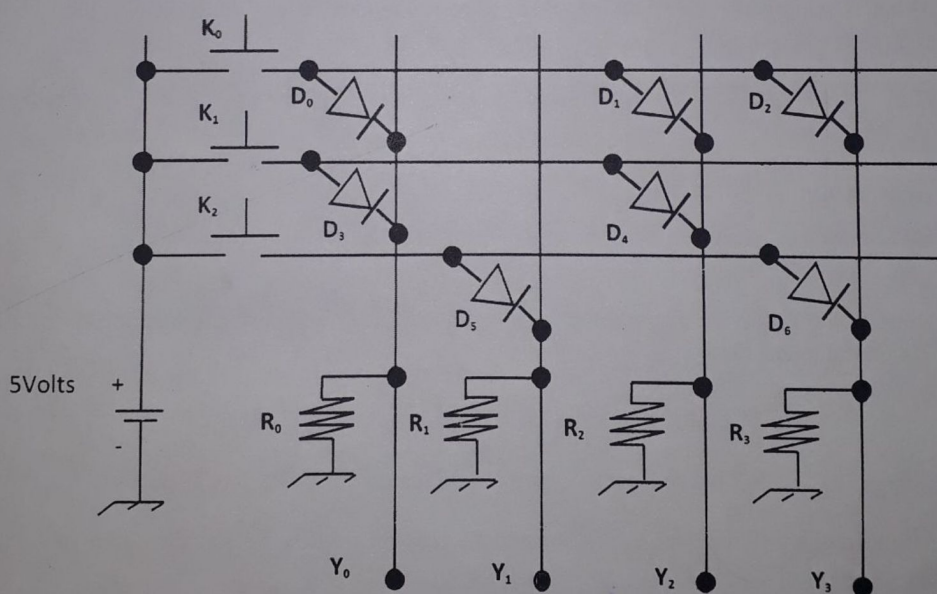
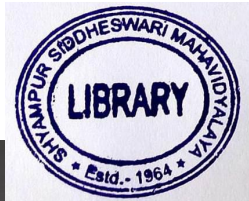
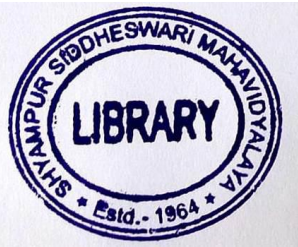


Figure (1)

- (c) What is the 'race-around' condition in a J-K flip-flop?

5+3+2



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2018

COMPUTER SCIENCE – HONOURS

Paper : CC-1

Full Marks : 50

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

Answer **Question no. 1** and **any four** from the rest.

1. Answer **any five** questions from the following :

2×5

- (a) Find the maxterm of $f(x, y, z) = xy + yz + zx$.
- (b) Draw the truth table for : $(A'+B) \oplus (A+B')$.
- (c) Compare between positive and negative edge-triggered clock pulses.
- (d) Convert the number $(DEC.A)_{16}$ to decimal.
- (e) How many bits are required to represent decimal values ranging from -50 to $+50$?
- (f) What is weighted code? Differentiate between weighted codes and non-weighted codes.
- (g) Add : (i) $(35)_8 + (53)_8$; (ii) $(9F)_{16} + (BA)_{16}$.
- (h) Subtract $(0111)_2$ from $(0011)_2$ using 1's complement method and also using 2's complement method.

2. (a) Convert :

- (i) $(11001111.1010)_2 = (?)_8$
- (ii) $(3A.2F)_{16} = (?)_2$
- (iii) $(396.82)_{10} = (?)_2$

(b) Convert the following decimal numbers to signed binary and perform the addition/subtraction using 2's complement method.

- (i) $33 + 15$
- (ii) $56 - 27$

(c) Design Y_{diff} of a 3-bit full subtractor by NOR gates only.

3+3+4

3. (a) Minimize the following logic function and realize using logic gates having minimum number of levels.

$$f = \sum m(1, 3, 5, 8, 9, 11, 15) + \sum d(2, 13)$$

Draw the truth table.

(b) Make a 4-input NAND gate using 2-input NAND gates.

(c) Realize Exclusive-OR logic function using minimum number of two-input NAND gates only.

5+3+2

Please Turn Over